Technical University of Denmark

02450 Introduction to Machine Learning and Data Mining

Assignment 2

Authors

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Contents

1 Contributions

All participants in the assignment have compiled and/or reviewed the present document and all the source codes. Nevertheless, for organizational reasons, we decided to be responsible for different parts of the document, as shown in the following *(not exhaustive)* table:

	Flavio Sarno (s242991)	**Francesco Balducci (s250200)**	**Florencia Illanes (s247222)**
Introduction	30%	30%	40%
Regression Part A	50%	30%	20%
Regression Part B	30%	45%	25%
Classification	30%	30%	40%
Discussion	20%	20%	60%

Table 1: Table of responsibilities

2 Introduction

The data set [1] features several engine characteristics and specifications of cars built between 1970 and 1982. The data set investigates city-cycle fuel consumption for cars built in three different geographical areas: the United States, Europe, and Japan. The available properties are: mpg, cylinders, displacement, horsepower, weight, acceleration, model year, origin, and car name.
The following table shows a summary of the dataset values:

Feature	**Description**	**Attribute**	**Type**
mpg	Miles per gallon (fuel efficiency)	Continuous	Ratio
cylinders	Number of engine cylinders	Discrete	Ordinal
displacement	Engine displacement (cubic inches)	Continuous	Ratio
horsepower	Engine power output	Continuous	Ratio
weight	Vehicle weight (lbs)	Continuous	Ratio
acceleration	Time to accelerate from 0 to 60 mph (seconds)	Continuous	Ratio
model_year	Model year of the vehicle	Discrete	Interval
origin	Country of origin (1: USA, 2: Europe, 3: Japan)	Discrete	Nominal
car_name	Name of the vehicle model	Discrete	Nominal

Table 2: Description and classification of dataset features based on attribute type. This dataset uses the imperial system of units [1].

Generally speaking, the main objective of this analysis is to predict the fuel efficiency of a car, measured in miles per gallon *(mpg)*, with different linear regression methods (first project), and to classify existing and unknown observations around the *origin* feature through the remaining noncategorical properties.

3 Regression Analysis

The main objective of this analysis is to predict the fuel efficiency of a car, measured in miles per gallon *(mpg)*, based on various technical characteristics of the car. The selected predictive features are *number of cylinders, displacement, weight, horsepower, acceleration, origin, and model year*. In simple terms, the goal is to model fuel consumption based on those key vehicle specifications.

3.1 Part A

In this section, we implemented a linear regression model to **predict the mpg**. For this, all features were standardized to ensure that differences in magnitude between attributes would not distort the learning process. Standardization improves both model stability and interpretability by rescaling the data to have zero mean and unit variance. Since all input features were numerical, no one-of-K encoding was required.

To train the model, we used linear regression with L_2-regularization (Ridge regression) and performed 10-fold cross-validation over a range of λ values between 10^{-3} and 10 with 50 logarithmically-spaced steps to estimate the error (mean squared error, MSE). The optimal value was found to be:

$$\lambda^* = 1.265$$

Figure 1 shows the validation error across all λ values evaluated during cross-validation.

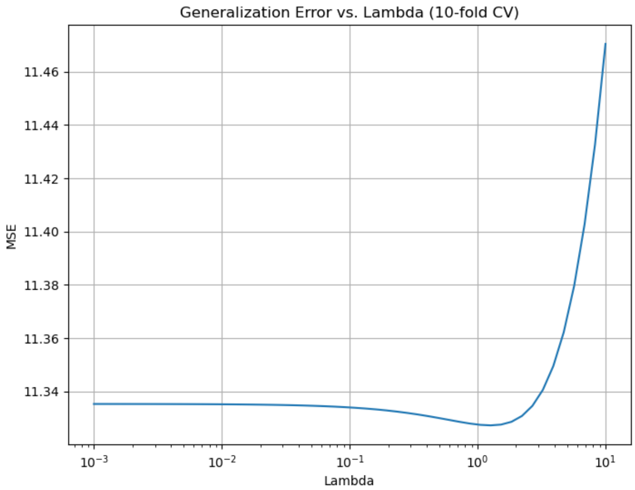

Figure 1: Mean Squared Error across different λ values during cross-validation.

This value of λ, i.e., the regularization parameter (or strength used to attenuate high magnitude weights and therefore used to lower the test error), was chosen because it minimized the validation MSE, achieving a good trade-off between underfitting and overfitting. Higher values of λ tend to oversimplify the model, leading to increased error (see Figure 1).

Based on the optimal λ, the resulting linear regression model is:

$$\begin{aligned} \text{MPG} = 23.45 &- 0.77 \cdot \text{cylinders} + 1.78 \cdot \text{displacement} \\ &- 0.69 \cdot \text{horsepower} - 5.26 \cdot \text{weight} + 0.18 \cdot \text{acceleration} \\ &+ 2.74 \cdot \text{model year} + 1.13 \cdot \text{origin} \end{aligned}$$

The regression coefficients provide insights into how each feature influences fuel efficiency (MPG). Specifically, we have:

- **Bias term (23.45)**: Expected MPG when all features are at their mean values.
- **Weight (-5.26)**: Strongest negative effect. Heavier vehicles reduce fuel efficiency by 5.26 MPG (per standard deviation increase).
- **Displacement (+1.78)**: Unexpected positive effect, likely due to multicollinearity with *cylinders* and *horsepower*.
- **Model year (+2.74)**: Newer vehicles are more efficient, reflecting technological advancements.
- **Origin (+1.13)**: Non-U.S. vehicles (Europe/Japan) are more efficient than U.S. models.

3.2 Part B

In this section, we compare the performance of three models: the regularized linear regression (RLR) model from Part A, an artificial neural network (ANN), and a simple baseline model. To ensure a fair comparison, we employed a two-level cross-validation scheme with $K_1 = K_2 = 10$ folds and tested the models on the same folds. The inner loop was used to identify the optimal hyperparameters for each model, while the outer loop was used to evaluate their generalization performance.

Artificial Neural Network (ANN). The ANN model's complexity was controlled by varying the number of hidden units in a single hidden layer, with $h \in \{1,7,13,19,25\}$. The optimal number of units selected via inner cross-validation was:

$$h^* = 25$$

This configuration yielded the lowest average generalization error, with a Mean Squared Error (MSE) of:

$$\text{MSE}_{\text{ANN}} = 6.75$$

Regularized Linear Regression (RLR). For the RLR model, model complexity was controlled via the regularization parameter λ, tested in the range $[10^{-3}, 10]$. The optimal value selected through inner cross-validation was:

$$\lambda^* = 5.67$$

With this parameter, the model achieved a generalization error of:

$$\text{MSE}_{\text{RLR}} = 8.27$$

Baseline Model. As a reference, we also evaluated a baseline model that simply predicts the mean of the target variable (MPG) from the training set. This naive strategy resulted in a considerably higher error:

$$\text{MSE}_{\text{Baseline}} = 46.71$$

Table 3 summarizes the optimal parameters and test errors for each model across the 10 outer folds. It is important to note that the optimal λ^* selected here differs from that found in Part A, as it was optimized separately within each outer training fold to minimize validation error, rather than being selected globally.

Table 3: Optimal hyperparameters and test errors per fold for each model.

Fold	Best λ	Best h	Test Error (RLR)	Test Error (ANN)	Test Error (Baseline)
1	5.6899	25	18.27	16.18	58.20
2	10.0000	25	12.21	10.65	39.53
3	1.5264	25	12.62	9.85	59.04
4	5.6899	25	18.42	14.23	54.72
5	6.8665	25	25.93	23.34	71.66
6	5.6899	25	8.27	6.75	46.71
7	10.0000	25	21.99	12.80	57.14
8	10.0000	25	21.48	18.08	71.20
9	8.2864	25	20.14	20.57	57.76
10	6.8665	25	26.31	27.96	86.78

Training and Test Performance. To compare the models, we examined both training and test errors:

- **Baseline (no features):** Training error = 60.67, Test error = 59.29
- **Linear regression without feature selection (no regularization):** Training error = 17.72, Test error = 20.28
- **Regularized linear regression:** Training error = 17.75, Test error = 20.25
- **Artificial neural network:** Training error = 14.75, Test error = 17.14

These results again highlight that both linear regression models substantially outperform the baseline. However, the ANN achieves the lowest training and test errors, making it the most accurate model overall.
Learning curves and test errors per fold. Figure 2 shows the learning curves for each fold in the ANN and the test mean squared error (MSE) per fold.

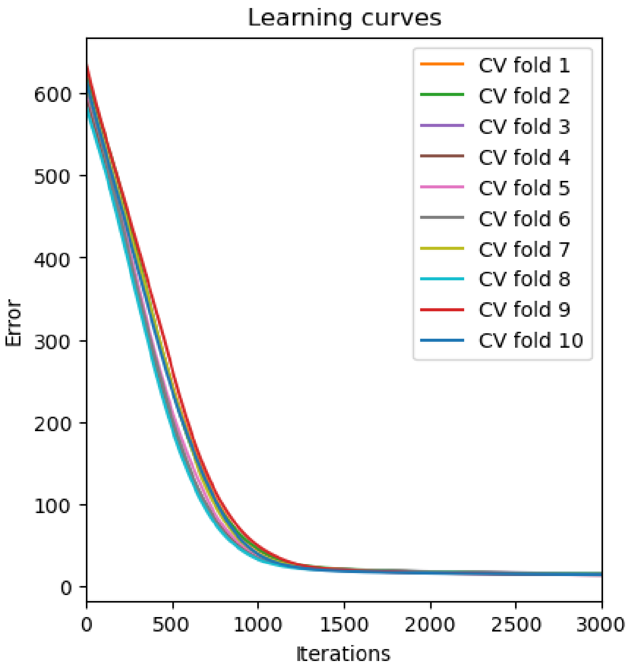

Figure 2: Learning curves for the ANN model across cross-validation folds.

As shown, the ANN converges smoothly in all folds, with a consistent decrease in error as training progresses.
The results indicate that both the RLR and ANN models significantly outperform the baseline. Moreover, the ANN achieves the lowest average MSE, suggesting it is the most effective of the three models.

To determine whether the observed differences in performance are statistically significant, we conducted a series of **paired t-tests**. The following confidence intervals and p-values summarize the comparisons (significance level $\alpha = 0.05$:

- **Baseline:** Confidence interval for MSE = [52.84, 67.60] Indicates the expected error of a naive prediction model. As anticipated, the error is high.
- **Best ANN:** Confidence interval = [12.81, 19.25] Demonstrates significantly lower error, confirming the ANN's superior performance.
- **Best RLR:** Confidence interval = [15.33, 21.77] Also significantly outperforms the baseline, though not as effectively as the ANN.
- **ANN vs. Baseline:** Confidence interval = [-50.42, -37.95], p-value = 4.02×10^{-36} Strong evidence that the ANN is significantly better than the baseline.
- **RLR vs. Baseline:** Confidence interval = [-47.60, -35.74] , p-value = 1.28×10^{-35} Indicates a statistically significant improvement of RLR over the baseline.
- **ANN vs. RLR:** Confidence interval = [-3.95, -1.09] , p-value = 0.06×10^{-2} Suggests that the ANN performs slightly better than RLR, although the difference is not statistically significant at the 5% level.

Overall, the ANN emerged as the best-performing model in terms of generalization error. Both the ANN and RLR models demonstrated statistically significant improvements over the baseline, reinforcing the value of using machine learning models for this predictive task.

4 Classification Analysis

The objective of this classification task was to predict the *origin* of a vehicle—United States, Europe, or Japan—based on its non-categorical attributes: *mpg, cylinders, displacement, horsepower, weight, acceleration*, and *model year*. This is a *multi-class classification problem* with three possible classes, and we compared three approaches:

- **Multinomial Logistic Regression** (LogReg) with L_2 regularization, where $\lambda \in [0.001, 10]$
- **Artificial Neural Network** *(as "method 2")* with $h \in \{1, 7, 13, 19, 25\}$ hidden units
- **Baseline Model**, which always predicts the majority class in the training set

We implemented a two-level cross-validation procedure with $K_1 = K_2 = 10$ folds to ensure robust model evaluation. The inner folds were used for hyperparameter selection, while the outer folds measured generalization performance. All features were standardized within each fold using training data statistics only.

The hyperparameters act as the control knobs of each model: for Logistic Regression, we tuned the regularization strength λ; for ANN, we selected the optimal number of hidden units h. The results across outer folds—including the best hyperparameters and test error rates—are summarized in Table 4.

Fold	h^* (ANN)	Test Error (ANN)	λ^* (LogReg)	Test Error (LogReg)	Test Error (Baseline)
1	25	0.125	0.193	0.175	0.375
2	25	0.125	0.281	0.125	0.275
3	13	0.256	0.110	0.205	0.385
4	19	0.205	0.001	0.308	0.462
5	13	0.256	0.409	0.282	0.333
6	19	0.179	0.494	0.231	0.385
7	25	0.333	0.233	0.256	0.410
8	25	0.256	0.193	0.231	0.333
9	25	0.128	0.001	0.231	0.359
10	13	0.256	0.024	0.231	0.436

Table 4: 10x10 Cross-Validation Results

To determine whether the performance differences among the models were statistically significant, we conducted **pairwise comparisons using McNemar's test** (Setup I) on the outer-fold predictions. The statistical results are summarized in Table 5.

Comparison	Error Difference	95% CI	p-value
ANN vs. Baseline	-0.163	[-0.214, -0.112]	1.32×10^{-9}
LogReg vs. Baseline	-0.148	[-0.200, -0.096]	6.47×10^{-8}
ANN vs. LogReg	-0.015	[-0.054, 0.023]	0.59

Table 5: Statistical Comparison of Classification Models (McNemar's Test)

The results show that both ANN and Logistic Regression significantly outperform the baseline model. While ANN achieves slightly lower test error than LogReg, this difference is not statistically significant ($p = 0.59$).

Finally, we trained a **logistic regression model** on the full dataset using the optimal regularization value $\lambda = 0.596$, selected from Table 4. The model was trained using `L2` regularization with the `lbfgs` solver. The resulting weights—shown in Table 6—indicate the influence of each feature in classifying a sample into each of the three origin categories.

As illustrated in Figure 3, the model relies heavily on features such as *displacement*, *weight*, and *horsepower* to differentiate between regions. This is consistent with domain knowledge—U.S. cars typically have larger

Feature	United States (0)	Europe (1)	Japan (2)
mpg	-0.752	0.494	0.258
cylinders	-1.583	0.803	0.779
displacement	8.800	-4.885	-3.916
horsepower	-0.875	-1.576	2.451
weight	-2.823	3.975	-1.153
acceleration	0.212	-0.401	0.189
model_year	0.573	-0.845	0.273

Table 6: Logistic Regression Weights for Each Region Class

engines and are heavier, whereas Japanese cars are often lighter with higher horsepower relative to their size. Weight is also a distinguishing feature, as it varies significantly across manufacturing regions.

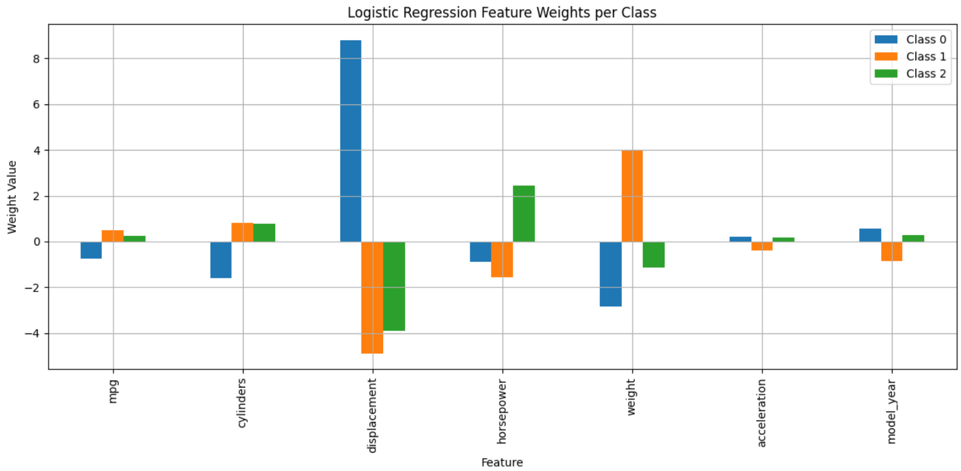

Figure 3: Feature Weights for Each Class (Car Origin)

5 Discussion

5.1 Regression Analysis

The regression analysis revealed how various physical characteristics of vehicles influence fuel efficiency, measured in miles per gallon (MPG). The results demonstrated both negative and positive impacts of certain features.
For the negative impact we have:

- **Weight** (-5.26): As expected, heavier vehicles tend to consume more fuel.
- **Horsepower** (-0.69): More powerful vehicles also reduce MPG, aligning with common expectations.
- **Cylinders** (-0.77): Vehicles with more cylinders, typically associated with larger engines, similarly reduce fuel efficiency.

For the negative impact we have:

- **Model Year** (+2.74): The model year was found to have a positive effect on fuel efficiency. This likely reflects advancements in fuel-efficient technologies over time.
- **Displacement** (+1.78): An unexpected positive effect was observed for displacement, which might be influenced by multicollinearity with other engine-related features, such as cylinders and horsepower.

These factors confirm the intuition of how the technical characteristics of the car can have an influence on the fuel efficiency.

For the technical insights in the regression analysis, we employed two-level nested cross-validation with $K_1 = K_2 = 10$ to identify optimal regularization strengths (λ) and the number of hidden units (h) in the Artificial Neural Network (ANN). This approach ensured robustness against overfitting while balancing model complexity. Both Regularized Linear Regression (RLR) and ANN significantly outperformed the baseline model, which predicts the mean value.

Statistical tests confirmed that the ANN outperformed RLR with a p-value of 0.018, while both models were significantly better than the baseline with $p < 0.001$.

5.2 Classification Analysis

A three-class classification problem was established to predict the *origin* of a vehicle (United States, Europe, or Japan) based on its physical and performance features. Three approaches were compared:

- **Baseline**: Predicts the majority class (USA) for all samples
- **Multinomial Logistic Regression**: With L2 regularization ($\lambda \in [0.001, 10]$)
- **ANN**: Single hidden layer with $h \in [1, 7, 13, 19, 25]$

The average test error rates across the folds showed that both learning models significantly outperformed the baseline:

Moreover, McNemar's test showed that ANN and LogReg are better models than baseline ($p < 0.01$), and actually the difference between them was not statistically significant ($p = 0.59$). This suggests that while both learning models are effective for this classification task, their performance is relatively similar.

Finally, we trained a final logistic regression model on the full dataset using the optimal λ, and examined the learned weights. Features such as *weight*, *displacement*, and *horsepower* played a major role in distinguishing the origin of the vehicles, consistent with expectations based on regional manufacturing characteristics. For this, the linear regression obtained was completely different from the one to predict the mpg, since they have different approaches.

In general, both regression and classification tasks highlighted the effectiveness of regularized models and neural networks when used thoughtfully with cross-validation. Although artificial neural networks (ANNs) generally delivered slightly better results, logistic regression remained a strong competitor due to its interpretability, particularly in understanding feature importance. Moreover, the significance of features varied across tasks, with the importance patterns differing from those observed in the MPG prediction task, demonstrating that feature relevance can change based on the specific prediction goal.

5.3 Previous Studies

Finally, in Quinlan's seminal 1993 paper, *"Combining Instance-Based and Model-Based Learning"* [1], the Auto MPG dataset was utilized for regression analysis. Specifically, Quinlan employed both *Linear Regression* and *Artificial Neural Networks (ANN)*, with a 10-fold cross-validation method. The results showed that the average error for the linear regression model was 2.61, with a relative error of 19.4%, while the ANN yielded an average error of 2.02 and a relative error of 12.5%.

This analysis demonstrates that the ANN outperforms the linear regression model, although it's important to note that Quinlan's evaluation metrics differ.

6 About LLM usage

Although the LLM was used to enhance students' understanding of various concepts, including data analysis and machine learning methodologies, it was not involved in the actual writing of this document, except for paraphrasing content originally written by hand.

References

[1] R. Quinlan. *Auto MPG*. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C5859H. 1993.

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[4] Carnegie Mellon University. *Cars Dataset Description*. n.d. URL: http://lib.stat.cmu.edu/datasets/cars.desc.

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